Affine ciphers.

An scheme of the form

$E(x) \equiv (ax+b) \bmod 26$ is called affine ciphir. Here x is the numerical equivalent of given Plaintext letter, a and b are integers.

Ex. Encipher ITS COOL with

$E(x) \equiv (5x+8) \bmod 26$.

Plain	I	T	S	C	O	O	L
x	8	19	18	2	14	14	11
$5x+8$	48	103	98	18	78	78	63
$(5x+8) \bmod 26$	22	25	20	18	0	0	11
ciphir	W	Z	U	S	A	A	L

If $y \equiv E(x) \equiv (ax+b) \bmod 26$ then we can solve x in tirm of y. So determine $E^{-1}(y)$. That is

$y \equiv (ax+b) \bmod 26$ then

$y-b \equiv ax \bmod 26$, or $ax \equiv (y-b) \bmod 26$

Let multiply both side by a^{-1} So

$x \equiv a^{-1}(y-b) \bmod 26$.

Hence $E^{-1}(y) \equiv a^{-1}(y-b) \bmod 26$.

Q- Decipher HPCCXD using.

$E(n) \equiv (5n + 8) \bmod 26$.

Sol.

$y \equiv E(n) \equiv (5n+8) \bmod 26$

$(y-8) \equiv 5n \bmod 26$

$5n \equiv (y-8) \bmod 26$

$5^{-1} \cdot 5n \equiv 5^{-1} \cdot (y-8) \bmod 26$

$n \equiv 5^{-1} \cdot (y-8) \bmod 26$.

Since.

$5 \cdot 21 \equiv 1 \bmod 26$

So $21 = 21$.

Use Extended Euclidean algo.

$\gcd(5, 26) = 1$.

$a \quad b$

$26 = 5 \times 5 + 1$

$5 = 1 \times 5 + 0$

$ax + by = \gcd(a, b)$

$5n + 26y = 1$

$1 = 26 - 5 \times 5$

$1 = 26 \times 1 + 5 \times (-5)$

Therefore -5 is inverse of 5.

$\hookrightarrow -5 + 26 \rightarrow 21$

Hence

$n \equiv 5^{-1} \cdot (y-8) \bmod 26$

$n \equiv 21 \cdot (y-8) \bmod 26$

ciphery	H	P	C	C	X	A	O
y	7	15	2	2	23	0	16
y-8	-1	7	-6	-6	15	-8	8
21(y-8)	-21	147	-126	-126	315	-168,	168
21(y-8) mod 26	5	17	4	4	3	14	12
Plain	F	R	E	E	D	O	M

Ex. Suppose an affine ciphery $E(x) \equiv (ax+b) \bmod 26$ encipher H as X and O as Y. Find the values of a and b and find encipher function.

Sol: $H \to X$ means $E(7) \equiv 23$ and $E(16) \equiv 24$.

$a \cdot 7 + b \equiv 23 \bmod 26$ and $a \cdot 16 + b \equiv 24 \bmod 26$

After Subtracting.

$16a - 7a \equiv 1 \bmod 26$ So $9a \equiv 1 \bmod 26$

$a \equiv 9^{-1} \bmod 26$.

Here $a = 3$ and $b = 2$

$E(x) \equiv (3x+2) \bmod 26$.